

Volume V · Chapter 23 — Robotics Fundamentals & Simulation (ROS 2, Isaac Sim) · Theory

Source: split from med-tech-curriculum-V1.md (V1). From this split onward this chapter is maintained **independently** here. See Volume-V-Chapter-23-context.md for the AI interaction log.

Prerequisites: Ch 4; general programming maturity.

Learning outcomes — the student can: explain robot kinematics/control basics; use ROS 2 concepts (nodes, topics, services, actions); run Gazebo/Isaac Sim; command a simulated arm.

Topics (theory) — 10 h:

#	Topic	h
23.1	Robotics basics: kinematics, DoF, coordinate frames	2
23.2	Control loops & real-time constraints	2
23.3	ROS 2 architecture: nodes, topics, services, actions	3
23.4	Simulation: Gazebo & NVIDIA Isaac Sim	2
23.5	Medical robotics landscape (surgical, assistive, lab automation)	1

Datasets/tools: ROS 2, Gazebo, Isaac Sim; GPU workstation. **Assessment:** simulation task deliverable (**60%**); ROS 2 quiz (**20%**); writeup (**20%**). **Key decisions:** simulator choice; **sim vs. hardware**; real-time constraints. **References:** plan §7; §13 → *Robotics & embodied AI*. **Hours:** Theory **10** + Lab **16** = **26**.